

ComfyUI Privacy & Safety — Student Handout

Running local AI video generation without leaking anything · verified Aug 2026

The 5 rules: ① use the **portable/standalone** version, not the Desktop app · ② treat **custom nodes** — not workflows — as the danger · ③ never launch with **--listen** · ④ avoid **API/cloud nodes** and never sign in · ⑤ work **offline** once models are downloaded. Follow these and your prompts, images, and machine stay private.

1 · Which version to install

- **Portable zip or git clone** (github.com/comfyanonymous/ComfyUI): **zero telemetry** — no analytics, no crash reporting, fully auditable source. It even disables HuggingFace's phone-home for you.
- **Desktop app:** has opt-in telemetry (device ID, usage events, crash reports). Policy says prompts/workflows/images are never collected — but a 2025 bug (fixed in v0.4.41) kept transmitting after users opted out. If you must use it: decline telemetry at first run and verify Settings → Telemetry is off.

2 · Custom nodes — the real threat (not telemetry)

Custom nodes are **arbitrary code running as you**. Every documented ComfyUI attack came through them: **ComfyUI_LLMVISION** (2024) stole passwords/wallets and enabled the 1.1 TB Disney breach; **Upscaler_4K** (2025) shipped a credential stealer *through the official registry* — ~779 infections before it was banned.

- Workflow files (.json) are just data — safe to read. The trap: loading one lists **missing nodes** and Manager offers to install them. That click is the attack surface. Vet the node list, not the JSON.
- Install the minimum. Prefer old, popular, actively maintained packs (Impact Pack, VideoHelperSuite, KJNodes). Distrust new packs with generic names ('upscaler', '4K'). Registry scanning helps but has demonstrably missed malware.
- Never run ComfyUI as admin. Keep crypto wallets / password managers off the machine that runs random nodes — or use a separate OS account.

3 · Network rules — understand the --listen flag

- Default is safe: ComfyUI listens on **127.0.0.1:8188**, meaning only programs on your own machine can connect. **--listen** rebinds it to every network interface, so any device on the same Wi-Fi can open it by browsing to your IP.
- **Why that's dangerous:** ComfyUI has no login and no API authentication — the network *is* the security boundary, and **--listen** removes it. Anyone on the Wi-Fi can queue workflows, read/write files, and install custom nodes (= run code on your machine). ~700 servers exposed this way were backdoored in 2025.
- **You can't flip it by accident — but it can be handed to you.** It only activates if the flag is in your launch command or .bat file. The realistic trap is copy-paste: YouTube/Reddit setup guides often include **--listen** without explaining it, and cloud GPU templates (RunPod etc.) ship with it preset. **Read every launch command before running it; if you see --listen, delete it.**
- Legit remote access (your desktop GPU from your laptop)? Don't use **--listen** — tunnel instead: `ssh -L 8188:localhost:8188 user@gpu-pc` then browse localhost:8188 on the laptop, or use Tailscale (free private VPN between only your own devices).
- Check yourself anytime: `netstat -ano | findstr 8188` should show only 127.0.0.1. Routine outbound traffic in a default install is just Manager registry fetches + update checks — set Manager **network_mode = offline** to silence them.

4 · Cloud / API nodes

Built-in API nodes (Kling, Veo, Sora...) send your prompt and images to comfy.org's servers *and* the vendor. Nothing is sent unless you use them — so don't: launch with **--disable-api-nodes** and never sign in to a Comfy account.

5 · Going fully offline

After models and nodes are downloaded, ComfyUI generates with **zero internet** — turn Wi-Fi off and verify yourself. First-run gotcha: some nodes fetch model files on first use, so run each workflow once before going offline.

Setup checklist ■ portable zip, not Desktop ■ launch flag: `--disable-api-nodes` ■ Manager `network_mode=offline` ■ Manager ≥ 3.38 ■ no `--listen` ■ nodes: minimal + reputable only ■ not running as admin ■ test with Wi-Fi off

Sources: comfy.org/privacy/desktop · github.com/Comfy-Org/ComfyUI SECURITY.md · blog.comfy.org Upscaler-4K post-mortem · docs.comfy.org partner-nodes · vpnMentor LLMVISION report · securityonline.info Pickai analysis